

ISLE OF PORTLAND, United Kingdom

WMO: 038570

Lat: 50.5218N

Lon: 2.4573W

Elev: 52

StdP: 100.70

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.5	0.8	-4.6	2.6	1.2	-3.2	2.9	2.1	20.4	8.4	18.6	8.3	7.8	50	0.782

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	21.3	17.3	20.1	16.8	19.1	16.3	18.1	20.1	17.5	19.2	17.0	18.4	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	18.8	16.9	12.1	18.1	16.4	11.8	17.8	51.3	20.2	49.5	19.2	48.0	18.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.5	14.1	-1.4	24.1	1.7	1.9	-2.6	25.5	-3.6	26.6	-4.6	27.7	-5.8	29.1		
(5)			DB	WB	-2.5	19.3	1.7	0.9	-3.7	20.0	-4.6	20.6	-5.6	21.1	-6.8	21.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	7.0	6.6	7.5	9.3	11.8	14.4	16.5	17.0	15.7	13.3	10.1	7.9
	DBStd	4.27	2.78	2.46	2.10	2.07	1.96	1.61	1.57	1.32	1.48	2.12	2.54	2.99
	HDD10.0	416	96	94	78	35	6	0	0	0	0	3	29	75
	HDD18.3	2521	352	327	336	272	203	119	61	46	80	155	246	324
	CDD10.0	946	2	1	1	13	61	132	201	217	171	107	33	9
	CDD18.3	9	0	0	0	0	0	1	4	4	1	0	0	0
	CDH23.3	8	0	0	0	0	0	1	4	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	7.4	8.9	8.1	7.4	6.5	6.4	6.0	6.0	6.3	7.1	8.3	8.5	8.9
Precipitation	PrecAvg	836	96	71	61	54	52	50	45	58	63	93	100	94
	PrecMax	1127	194	165	130	137	111	122	122	129	129	141	207	173
	PrecMin	667	19	11	3	2	8	8	6	9	13	33	24	29
	PrecStd	119	42	39	26	33	28	28	27	32	32	32	44	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	11.6	12.9	17.2	19.7	21.7	23.7	23.2	21.0	18.0	15.3	13.2
		MCWB	11.0	10.3	9.5	12.8	15.3	17.2	17.6	19.0	17.2	15.9	14.3	12.3
	2%	DB	11.4	10.7	11.3	14.7	17.3	19.3	21.4	21.1	19.4	17.1	14.5	12.7
		MCWB	10.5	9.8	9.5	11.1	14.0	15.7	17.2	17.6	16.5	15.4	13.5	11.9
	5%	DB	10.8	10.1	10.6	13.0	15.7	17.9	20.1	19.8	18.6	16.4	14.0	12.2
		MCWB	10.1	9.2	9.2	10.5	13.3	15.1	16.7	17.0	15.9	15.0	13.0	11.3
	10%	DB	10.3	9.6	10.0	11.8	14.6	16.8	19.0	19.0	17.8	15.9	13.4	11.6
		MCWB	9.5	8.7	8.9	9.8	12.6	14.6	16.3	16.5	15.6	14.6	12.3	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.0	10.7	13.4	16.5	17.6	18.6	19.6	18.1	16.7	14.8	12.9
		MCDB	11.7	11.1	11.6	16.4	18.6	20.6	21.7	22.3	19.8	17.4	14.9	13.0
	2%	WB	10.9	10.2	10.2	11.7	14.6	16.4	17.7	18.4	17.5	16.1	14.0	12.2
		MCDB	11.2	10.4	10.8	13.6	16.4	18.6	20.2	20.2	18.7	16.7	14.2	12.5
	5%	WB	10.3	9.6	9.7	10.9	13.7	15.6	17.2	17.7	16.9	15.6	13.3	11.6
		MCDB	10.6	9.9	10.3	12.3	15.2	17.3	19.2	19.1	17.9	16.2	13.8	12.1
	10%	WB	9.8	9.0	9.2	10.4	13.0	15.0	16.7	17.2	16.4	15.0	12.6	10.9
		MCDB	10.2	9.4	9.8	11.5	14.3	16.4	18.4	18.3	17.4	15.7	13.3	11.5
Mean Daily Temperature Range	5% DB	MDBR	2.9	3.2	3.6	4.1	4.2	4.1	4.0	3.9	4.0	3.3	3.2	3.0
		MCDBR	2.7	3.0	3.8	5.5	5.6	5.6	4.9	4.6	3.2	2.8	2.7	
		MCWBR	3.4	3.1	3.1	3.8	3.9	3.8	3.6	3.3	3.2	2.9	3.4	3.6
	5% WB	MCDBR	2.7	2.8	3.2	4.2	5.0	4.8	4.7	4.1	3.8	2.8	2.7	2.7
		MCWBR	3.5	3.2	3.0	3.3	3.7	3.5	3.3	3.1	3.1	3.1	3.5	3.6
Clear-Sky Solar Irradiance	taub		0.333	0.348	0.382	0.396	0.399	0.392	0.387	0.380	0.371	0.360	0.333	0.323
	taud		2.385	2.397	2.306	2.295	2.333	2.383	2.402	2.442	2.452	2.449	2.434	2.361
	Ebn at Noon		690	782	814	845	858	866	864	854	821	757	687	637
	Edh at Noon		65	82	106	119	120	115	111	102	91	77	61	57
All-Sky Solar Radiation	RadAvg		0.82	1.57	2.80	4.41	5.33	5.78	5.49	4.59	3.47	1.94	1.04	0.65
	RadStd		0.07	0.17	0.35	0.39	0.28	0.57	0.39	0.36	0.22	0.14	0.10	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page